

CLASS 06 / SEP 29 / U04

ARRAYS

1D · 2D · 3D

P3b BEGINS — SHEET 1: ONE MODULE, MANY POSITIONS

P3a DUE

P3b ISSUED TODAY / P2b PRINT CHECKPOINT TODAY

AN ARRAY IS A LIST OF POSITIONS

Not a command that duplicates things. A list you compute, then place your module at.

INDEX → POSITION

$i = 0 \rightarrow p = \text{start}$

$i = 1 \rightarrow p = \text{start} + \text{step}$

$i = 2 \rightarrow p = \text{start} + 2 \times \text{step}$

$p[i] = \text{start} + i \times \text{step}$

THREE THINGS TO READ

THE LIST — what is in the collection.

THE COUNT — how many there are.

THE INDEX — where each one sits in the order.

The index is the only new idea. It is a counter that tells each copy how far along it is — and therefore where it goes.

Your P3a module is the component. It does not change in P3b — only where it goes.

THE REPEAT ZONE

The loop, made visible. Repeat Input, a body, Repeat Output.

WHAT IT IS

Repeat Input → **body** → **Repeat Output**

Everything between the two runs once per iteration. Whatever you hand to Repeat Input comes back changed, once for every step of the count.

READ IT SMALL FIRST

Set the count to 3 and look at the spreadsheet.

Then 4. Then 5.

A collection you cannot inspect is a collection you cannot debug.
There is no minimum count and no prize for a large one.

BUILD IT AT A COUNT YOU CAN COUNT. SCALE UP ONLY ONCE IT IS RIGHT.

ONE COUNTER, TWO, THREE

Linear, nested grid, XYZ cube. Each one is the last one, repeated along a new axis.

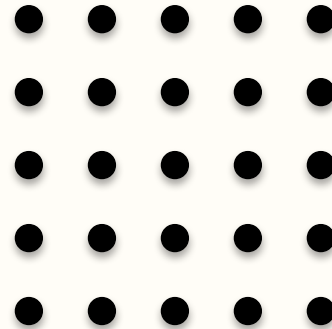
1D — LINEAR



$$\text{count} = n$$

$$n = 5 \rightarrow 5$$

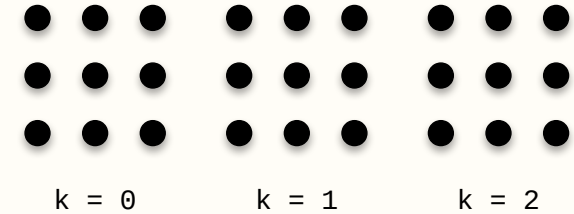
2D — NESTED GRID



$$\text{count} = n \times n$$

$$n = 5 \rightarrow 25$$

3D — XYZ CUBE



$$\text{count} = n^3$$

$$n = 3 \rightarrow 27$$

PREDICT THE TOTAL BEFORE YOU LOOK. COMPARE 3, 4 AND 5 PER AXIS: 27, 64, 125.

INDEX ORDER

Which counter runs inside decides the order. Label it on the sheet.

A 4 × 3 GRID, i INSIDE j

8	9	10	11	j = 2
4	5	6	7	j = 1
0	1	2	3	j = 0
i → 0 · 1 · 2 · 3				

THE FLAT INDEX

2D: $\text{index} = i + j \times \text{cols}$

3D: $\text{index} = i + j \times n + k \times n^2$

Swap the loops and the same positions get different numbers.
The picture looks the same; any rule that reads the index does not.

SAY IT OUT LOUD: WHICH WAY DOES i RUN, AND WHAT HAPPENS IF YOU SWAP THE LOOPS?

COMPARE + SWITCH IS AN IF STATEMENT

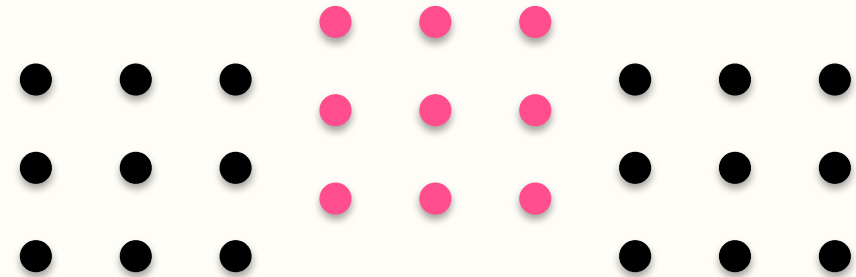
Test the index, then act on the answer. One conditional selection is encouraged in P3b.

THE PATTERN

Compare $3 \leq i \leq 5$ $\rightarrow$ True / False
Switch False $\rightarrow$ stay
 True $\rightarrow$ lift

Compare asks the question. Switch chooses between two answers.
Everything conditional in this course is those two nodes.

AN INDEX RANGE, LIFTED



Indices 3 to 5 moved. The module is the same; only its placement changed.

CHECK THE RESULT AT TWO DIFFERENT COUNTS. IF IT ONLY WORKS AT ONE, IT IS NOT A RULE.

SHEET 1, STEP BY STEP

What we build together today: 1D, 2D, 3D, then one conditional.

01

Week 6 · 1D: build a small linear array with Repeat Input and Repeat Output. Identify list, count and index.

02

Week 6 · 2D: nest repetition into a grid. Identify rows and columns; predict the total count.

03

Week 6 · 3D: repeat the grid in Z for a cube array. Predict Count cubed; distinguish instances from a sampled volume.

04

Use Compare and Switch for an if condition; select an index range and move only those copies.

P2b CHECKPOINT — PRINT GEOMETRY

Check geometry, units, orientation and the sliced preview before queueing.

GEOMETRY

Closed and solid. Every edge shared by two faces, no stray pieces.

UNITS

Real millimetres. Within 60 mm on each axis.

ORIENTATION

No supports. Orientation is your lever; change the geometry if you must.

SLICED PREVIEW

Read it layer by layer before queueing. Stay within the 200 g allocation.

P2b IS DUE AT THE MIDTERM, WEEK 10 — OCT 27. BRING THE PRINT AND ITS PROCESS SHEET.

ISSUED TODAY

P3b — ARRAYS

DUE WEEK 8 — Oct 13 / 7%

One assignment, exactly two 17×11 sheets, submitted together. Array your P3a module without changing it: sheet 1 today, sheet 2 next week.

SHEET 1 · TODAY

1D · 2D · 3D

Linear, nested grid, XYZ cube. Count, spacing, index order and predicted totals labelled.

SHEET 2 · WEEK 7

HEX · RADIAL · CURVE

Staggered grid, radial array, and an array along a sine-driven curve.

THE RULE

THE MODULE STAYS

Vary only its placement. Expose count and spacing or radius in the graph.

REQUIREMENTS — P3b

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Array your P3a module across exactly two 17×11 inch sheets, submitted together as one assignment.

02

Sheet 1 — 1D, 2D and 3D arrays: a linear array, a nested grid and an XYZ cube array. Label count, spacing and index order; show the predicted total for each.

03

Sheet 2 — hexagonal, radial and curve arrays: a staggered hexagonal grid, a radial array (radius, count, angular step) and an array along a curve, including one sine-driven curve with amplitude, frequency and phase labelled.

04

Keep the module unchanged; vary only its placement. Expose count and spacing or radius in the graph.

05

Build repetition from basic nodes and begin with a count you can inspect; there is no minimum element count. One conditional (index-range) selection is encouraged on either sheet.

NOW — SHEET 1

Rest of the class: three arrays you can predict, and one conditional.

✓

Your module on a linear array, count and spacing exposed, spreadsheet open

✓

The array nested into a grid, total predicted before you looked

✓

The grid repeated in Z: 3, 4 and 5 per axis compared

✓

Index order labelled, and one index range moved with Compare and Switch

✓

P3a submitted, and your P2b print geometry checked

A COUNT YOU CAN COUNT. NOT A THOUSAND OF ANYTHING, YET.

BEFORE NEXT CLASS

NEXT

Week 7: hexagonal, radial and curve arrays — sheet 2 of P3b. Bring sheet 1's graph working; we build on it.

DUE

Nothing is due next week; it is a studio-review week. P3b, both sheets together, is DUE WEEK 8 — Oct 13 / 7%.

BUILD

Finish sheet 1: linear, nested grid and XYZ cube, with count, spacing, index order and predicted totals labelled.

PRINT

P2b: fix anything flagged at today's checkpoint, then queue it. It is due at the midterm, WEEK 10 — Oct 27.